# Identifying Onboarding Heuristics for Free-to-Play Mobile Games: A Mixed Methods Approach

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**Abstract**
The onboarding phase of Free-to-Play mobile games, covering the first few minutes of play, typically sees a substantial retention rate among the players. It is therefore crucial to the success of these games that the onboarding phase promotes engagement to the widest degree possible. In this paper a set of heuristics for the design of onboarding phases in Free-to-Play mobile games is presented. The heuristics are identified by a lab-based mixed-methods experiment, utilizing lightweight psycho-physiological measures (Galvanic Skin Response and Photoplethysmography), two questionnaires, post-session interviews, stimulated recall and engagement graphs, across three titles that cross the genres of puzzle games, base builders and arcade games, and utilize different approaches to onboarding phase design: *Candy Crush Jelly Saga*, *WinterForts* and *Pogo Chick*. Results showcase how heuristics can be used to design engaging onboarding phases in mobile games, and provide support for the use of lightweight psycho-physiological methods in mobile game evaluation.

**Keywords:** Mobile Games • Flow • Psycho-physiological Measures • User Experience • Heuristics • Onboarding Phase • Free to Play • Game User Research

## 1 Introduction

The Free-to-Play (F2P) business model has become increasingly popular when it comes to mobile games, and currently dominates the market. However, thousands of F2P mobile games exist, suffering from discoverability issues, and are characterized by imbalances in player retention[1]. Appsee [2] noted that the retention rate of mobile games is relatively low, 28.60% of players return to the game after one day[2]. The reasons for the typical low retention rates varies, but one out of the four most common is a poorly designed onboarding experiences [2]. The onboarding phase is thus an important element for capturing and retaining new players but to the best knowledge of the authors, no heuristics have yet been developed for purpose of designing the onboarding phase. This paper will therefore focus on developing onboarding heuristics for mobile games in order to provide the industry with a set of guidelines worth considering when designing a mobile game. The exact duration of the onboarding phase of a F2P mobile game varies across titles, but in collaboration with the developers at King and Norsfell,

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seven minutes were established as a generic timeframe among the evaluated games. Broadly speaking, the onboarding phase lasts from the first time a player starts the game until the basic mechanics have been utilized.

The heuristics presented in this paper are formed based on an experimental study that evaluated three mobile titles: *Candy Crush Jelly Saga* (King, 2016), *WinterForts* (Norsfell, 2015) and *Pogo Chick* (Norsfell,2016). Two lightweight psycho-physiological measures, GSR and PPG were combined with abbreviated versions of the Flow State Scale [3], using this as a proxy measure of engagement, as well as post-session walkthroughs and user-generated engagement graphs. These data were combined to develop an understanding of the features of the three games which influence flow and user experience, and jointly permitted the definition of heuristics that focus on fostering engagement of players during the onboarding phase.

**1)** *Candy Crush Jelly Saga* is a candy themed puzzle game where the game objective is to match three or more pieces of candy in the same colour, in order to remove jelly and hereby win the levels. **2)** *WinterForts: Exiled Kingdom* a strategy and base builder game where the game objective is to build, maintain, construct alliances and attack enemies. **3)** *Pogo Chick* is an arcade-style game where the player controls a chick on a pogo stick through a track of obstacles. The games objective is to transport the chick as far as possible without falling over. The onboarding phases of the three titles are highly diverse: *Candy Crush Jelly Saga* is structures by levels but provides perceived autonomy, *Winter Forts* is highly structured and all players will have to perform the same task complete it and *Pogo Chick* only provides signifiers of thumbs to indicate the games controls.

## Related work

The related work section will focus on key related work across heuristics in games, psycho-physiological methods in game users research and the use of Flow theory in game evaluation. **Heuristics in games:** Within the context of digital games, multiple sets of heuristics have been developed and related studies suggest that heuristics are excellent at identifying playability issues in prototypes [4] which makes them a valuable tool for the early development phases of games. However, during a comparison of two set of playability heuristics, Korhonen et. al [4] found that the amount of heuristics should not be too vast as redundancy then might occur between heuristics and that they should be presented similarly to Nielsen's ten usability heuristics [5, 4]. **Psycho-physiological measures in games:** The domain of psycho-physiological research has been described as being concerned with trying to decipher mental processes, by measuring the body's signals and thereby finding links between them [6]. Emotions are a core element in experience and using psycho-physiological measures to recognizing a user's emotional state, is utilized across fields that share related psycho-physiological states, e.g. arousal [7]. While a full breakdown of all relevant related work on psycho-physiological measures in games user research is out of scope here, a mention is here made of work that utilizes the same measures in surrounding fields. In a similar study, also using a Cinema setup, [8] used ECG and GSR to measure implicit psychological constructs and found physiological observations are better suited for observation of e.g. archetypes, than introspective reports. PPG is like ECG a measure that equally can be used to calculate an HRV score, which is a widely used scale to measure arousal. However,

determining whether a psycho-physiological response is related to an event or non-specific can be a challenge because the many-to-one relationship plays an important role [9]. **Flow:** Flow theory was developed by Csikszentmihalyi [10] based on interviews and observations of e.g. chess players, musicians and athletes. The focus of the work, is the relationship between intrinsically motivated behavior and the associated satisfaction from performing actions in their own right, as compared to the accomplishing of a goal. According to Csikszentmihalyi the concepts of flow consist of nine dimensions: Challenge-Skill Balance, Action-Awareness Merging, Clear Goals, Unambiguous Feedback, Concentration on Task at Hand, Sense of Control, Loss of Self-Consciousness, Transformation of Time and Autotelic Experience [10]. These nine dimensions are adopted in the Flow State Scale (FSS), originally developed for measuring Flow in sports [3], but adopted for use in digital games Kivikangas and further validated by Nacke et al. [11]. The FSS measures the nine dimensions of flow, via 36 questions, 4 per dimension. Related surveys developed specifically for games, are targeting the user experience more broadly than Flow but still includes elements of the nine dimensions. The *Game Experience Questionnaire* [12] and *The Game Engagement Questionnaire* [13] (both in literature abbreviated GEQ) are psychometric instruments for game user experience assessment. However, the FSS is the only frequently used questionnaire utilized for the evaluation of flow experience [11], and thus adopted here. A short versions of the *Game Experience Questionnaire*, termed the *Post-Game Experience Questionnaire* (PGQ) [12], covers the user experience dimensions of Positive and Negative effect, Tiredness and the feeling of Returning From a Journey. The PGQ was adopted here as a proxy measure of the user experience.

## 2 Method

In brief, the participants played the onboarding phase of three different mobile games, which possess different onboarding styles. The participants played the onboarding phase of each game, without prior experience with the games. GSR and HRV data was collected during the play session, while questionnaires, interviews and engagement graphs data was collected post play. Psycho-physiological measurements measures were used as the interest was in knowing if these could be used to evaluate engagement and experience of mobile games. The experimental setup was based on a within-subject design where each participant was exposed to three different onboarding phases.

The participants included 21 females (75%) and 7 males (25%). The age of the participants ranged from 20-37 years (mean = 25.25; SD = 3.63). Demographic data was collected as well as information on prior game experiences.

Skin conduction levels were measured using a Bitalino GSR sensor using two pre-gelled Skintact F-401C Ag-AgCl electrodes. The electrodes were attached to the participant's ring and little finger on the non-dominant hand. IBI was recorded by a Merlin-digital Heart Rate Monitor. The PPG sensor was attached to the participants left ear lobe. An IPad mini 3 was used for the test sessions. The IPad was placed in a stable mount in front of the participants. Two cameras where used one recording the participants and the other recording the screen of the IPad. Before each play session a three-

minute baseline was recorded, where after the participants played the onboarding phase of a game. The post-play sessions consisted of multiple steps: a) drawing first engagement Graphs; b) FSS questionnaire; c) stimulated recall; d) drawing second engagement graph; e) PGQ questionnaire. This approach was followed for each onboarding phase condition. Before the first test session pilot testing was conducted to validate the setup and detect problems.

## 3 Data analysis

Data analysis consisted of multiple steps: a) post-session processing of the psychophysiological data; b) evaluation of the FSS and PGQ surveys; c) explorative content coding of the post-session interviews; d) evaluation and comparison of the engagement graphs with GSR and HRV; e) comparative and correlational analysis across the qualitative and quantitative measures. The GSR data was first visually inspected for logging gabs, where after the data was normalized using an algorithm that was used in similar analysis by Mirza-Babaei et al. [14]:

$$GSR_{Normalized} = \frac{(GSR_{(t)} - GSR_{Min})}{(GSR_{Max} - GSR_{Min})} \quad (1)$$

*GSRMin and GSRMax* refers to the minimum and maximum GSR value in a certain time frame and. *GSRt* is the GSR data point in the middle of the time frame [14].

The GSR and HRV data was divided into smaller time windows that were based on the game design. Based on these time frames, graphs were created to visualize the normalized GSR and HRV scores. The GSR and HRV graphs were compared to the engagement graphs, created by the game designers and participants. Only a visual comparison was performed in relation to the four graph types, due to the subjective nature of the engagement graphs.

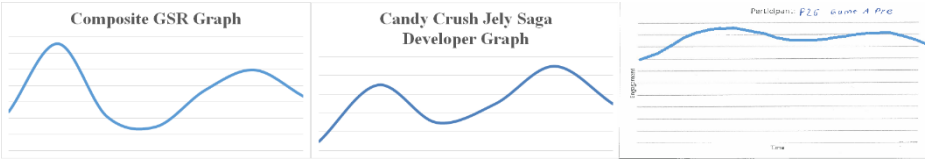


**Fig. 1.** Example of the four different graph types, Composite GSR(Left), Developer Engagement Graph (Middle), Player drawn Engagement Graph (Right).

In order to check the reliability of the PGQ and FSS, Cronbach's alpha was calculated. For the FSS the average score was .64 (SD = .42) while the PGQ had an average score of .71 (SD = .13). In order to investigate which of the three games induced the highest flow score, the average score across all flow dimensions was calculated. A comparison of overall flow scores is displayed in Table 1.

**Table 1.** Average FSS score for the three tested games and the PGQ dimension positive and negative experience

| Game | Average flow scores | PGQ Positive Dimension | PGQ Negative Dimension |
|---|---|---|---|
| *Candy Crush Jelly Saga* | 3.84 (SD=1.03) | 2.27 (SD=1.19) | 1.52 (SD=.94) |
| *WinterForts* | 3.11 (SD=1.26) | 1.74 (SD=.95) | 2.04 (SD=1.40) |
| *Pogo Chick* | 3.04 (SD=1.30) | 2.08 (SD=1.40) | 1.83 (SD=.99) |

The coded interview data showed that one of the main problems with *WinterForts* is missing autonomy, 75% of the participants named this during the interviews. While *Pogo Chick* was rated negatively because of the difficulty (intended high by designers) which 50% of the participants stated.

The video recordings were meaning condensed and then coded by the means of an open coding. The engagement graphs were further analyzed by comparing them to the statements given during the interviews. The sentiments describing the participant's experience of the onboarding phase and thus the graphs, were used to analyze which game element created an enjoyable onboarding phase. Based on the analyzed data several onboarding heuristics were identified.

## 4 Identifying Onboarding Heuristics

The following top level categories were identified for the open coding: 1) Game context, 2) Music and 3) Learnability, 4) Rewards System 5) Relatedness 6) Mastering Controls 7) Game Objective and 8) Progress Overview. The categories 1,2 and 3 seemed to be related to the device being mobile and the context thus ever-changing. The players were e.g. conscious of not disturbing their surrounding while playing and also needed games that can be played despite of interruptions caused by their surroundings. 4,5 and 8 are related to the games providing valuable and meaningful feedback for the players in order for the players to be able to relate to a game they never played before. Lastly, 6 and 7 is related to the game mechanics, where the controls need to be manageable and the games objective need to be clear to the player. Additionally, several of the categories identified (4, 6, 7 and 8) in the open coding showed a shared relationship with the nine dimensions of Flow and a selective coding was therefore performed to investigate if more categories would uncover. The selective coding uncovered the supplementary categories Immersion and Autonomy. The categories 6, 7 and 8, initially found during the open coding, were refined and renamed: Skill Level, Clear goal and Clear feedback and Progression.

## 5 Results

The heuristics identified in the empirical data will know be presented. The description of each heuristic follows the style of Nielsen's [5], by promoting genericity in the limited amount and is easier to manage when applied. The heuristics are: **Clear goal:** Players should have a clear understanding of the games goal; this can be established in the

beginning or during gameplay. Clear goals are important for the players to see purpose in their actions. **Autonomy:** Players should feel in control when playing a game. Missing autonomy can cause a negative game experience. Lack in autonomy can result in the players feeling passive and even stupid. **Skill Level:** The game should match a player's skill, without being too hard or easy. players will have difficulty reaching a state of immersion, if the game is either taking too much or too little cognitive effort. **Relatedness:** Players need to feel an association with the game. The player thus needs to feel that e.g. the theme, visual design and genre of the game is relatable and has a connection with their understanding of themselves. **Clear feedback and progression:** Players should receive information about how well they are during and when succeeding in a goal. Too much information can overwhelm players while to little can confuse them. **Game context:** The game needs to allow varying contexts and for disturbances without the players making irreversible mistakes that permanently ruins the player game experience. **Learnability:** Players should be able to learn core game mechanics quickly during the onboarding phase, to ensure that they know how to play the game. Players should also be able to quite the game during the learning phase. **Music:** Music in the game can help support the game genre and environment but usages of sound depends on the surrounding context, as the players can be cautious about causing disturbance. Game should thus be enjoyable without sound. **Rewards system:** Rewards needs to have value and purpose for the player. Through the onboarding phase the player needs to learn the reward system and find the rewards desirable to obtain when playing. **Immersion:** Immersion can be the outcome of a well-designed onboarding phase. The game needs to be challenging the player but not be stressful, it needs to provide a clear goal whiteout inhibiting the feeling of autonomy, it needs to provide the player with progression and feedback.

## 6 Discussion and Future work

In this paper the result from a mixed-methods empirical study focusing on investigating the onboarding phase of F2P mobile games have been reported in the form of a set of design heuristics. The goals of this study was to identify which elements that create an enjoyable onboarding phase, and based on this identify heuristics aimed towards assisting designers. Furthermore, lightweight psycho-physiological measurements (GSR and PPG) were utilized to investigate if it was possible to utilize the techniques in a mobile game setting. It was concluded that these measures of arousal appear to correlate with the reported experience of the players and that the high-frequency measures can be used to obtain insights into the specific causes of arousal. This result parallels previous work in other types of games [11, 6]. In this study the onboarding phases of three mobile games were inspected, and several problems were located which caused disengagement among some of the participants. During the evaluation of the three games, the most common problem identified during the onboarding phase was that players did not feel in control during the tutorial phase. When players were forced to perform certain actions, without the possibility of choice the participants in this study explained a negative game experience. The heuristics presented here were developed based on empirical

findings, identified during the evaluation, and theoretical concepts. However, they cover similar areas of interest with related literature on the development of playability heuristics, which indicates that the heuristics, presented in this paper, might be more generic across games and platform. Several of the identified heuristics have also been described in other studies [15, 16], which includes: Clear Goal, Autonomy, Skill level, Clear Feedback and Progressions and Reward System. Even though several sets of heuristics have been developed, none of them are specially designed to improve the onboarding phase of mobile games. The onboarding heuristics thus contributes with knowledge within a relatively undiscovered research area of mobile games. As the presented heuristics still are at an early stage in their development, they will now have to be tested an evaluated. Which our future work will focus on. By testing and evaluating the heuristics it should be possible to create more refined and effective heuristics, which game developers can use to create better onboarding phases and thereby increase the player's game experience and retention rate. Additionally, this study indicated that lightweight psycho-physiological and self-reported data can support each other in the evaluation of the onboarding phase of mobiles games. Lightweight psycho-physiological measures can pinpoint even small game events that causes high or low arousal, which could be investigated and elaborated further in the future. In conclusion, the three tested games had very different onboarding phases, which helped identifying the elements that created a more or less enjoyable experience for the participants. This formed the basis for identifying the 10 onboarding design heuristics. This study also indicated that lightweight psycho-physiological measures could be used to evaluate the onboarding phase of mobile games. However, future research has to be conducted to provide deeper insight into the use of psych-physiological measures and the developed heuristics.

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